UNIVERSITY OF RWANDA
COLLEGE OF SCIENCE & TECHNOLOGY
SCHOOL OF ICT
DEPARTMENT OF COMPUTER ENGINEERING
MODULE: MOBILE APPLICATION SYSTEMS & DESIGN
REG NO: 223007703

Q1. Dependencies used & why SQLite.

=> In my flutter project I used the following packages in pubspec.yaml.

- sqflite : ^2.3.0 => SQLite database engine for flutter (Android/ios/Desktop).
- path : ^1.8.3 => Cross-platform path joining to locate the DB file on disk
- path_provider : ^2.1.1 => Accesses the correct device file system directory for the DB
- intl : ^0.19.0 => Formats DateTime objects for display in the UI.

Q1.2. Why SQLite is Necessary for local storage.

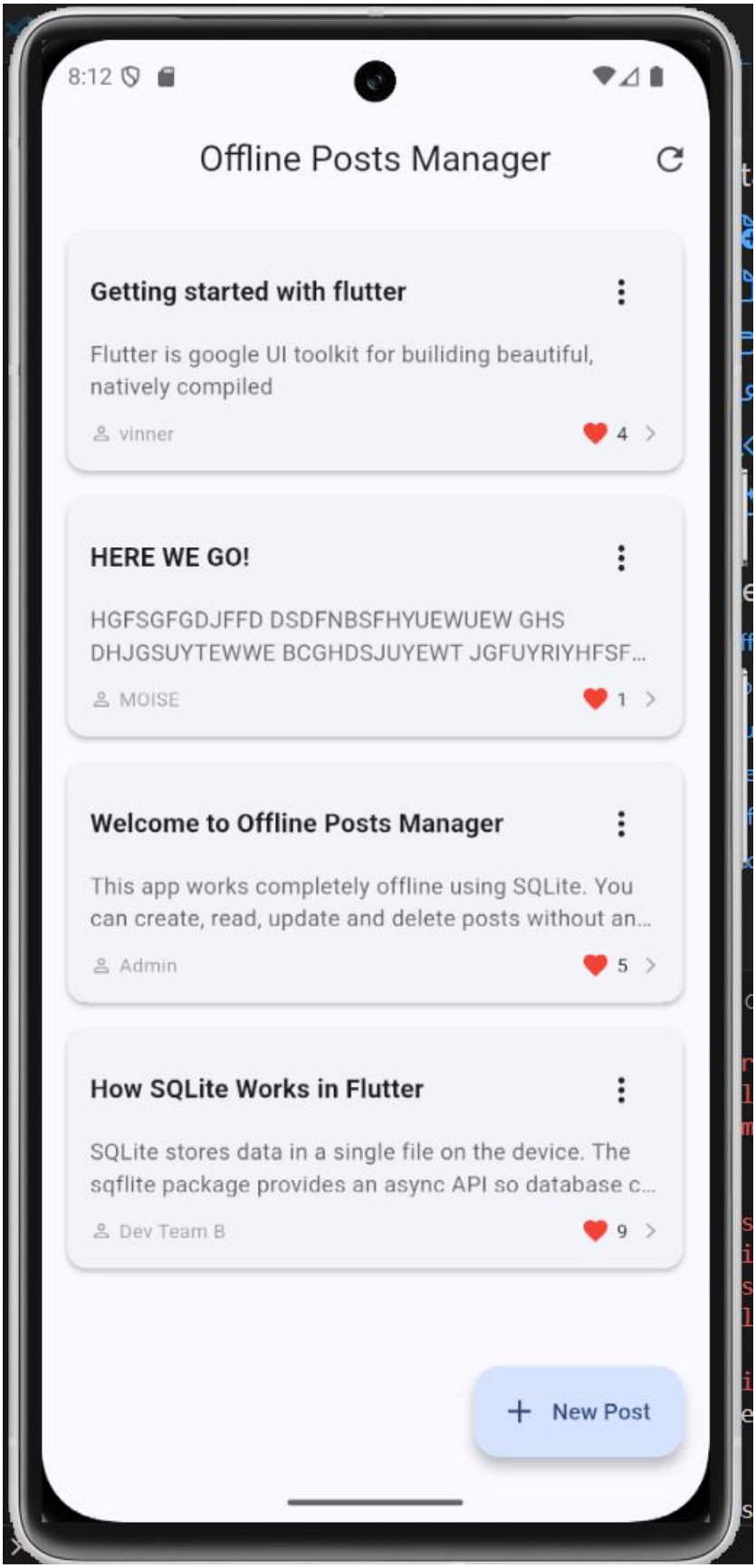
- SQLite is an embended relational database. It runs inside the app process with no server required.
- Data is stored in a single .db file on the device, so it persits across app restarts without any internet connection.
- It supports full SQL (select, insert, update, delete) making structured queries, filtering and sorting straightforward.
- Alternatives like sharedPreferences only store simple key-value pairs - unsuitable for relational data (posts with replies, likes, authors, timestamps).
- SQLite is ACID-Compliant, meaning write are atomic and the database stays consistent. even if the app crashes mid-write.

Q2. Handle Database Not initialized.

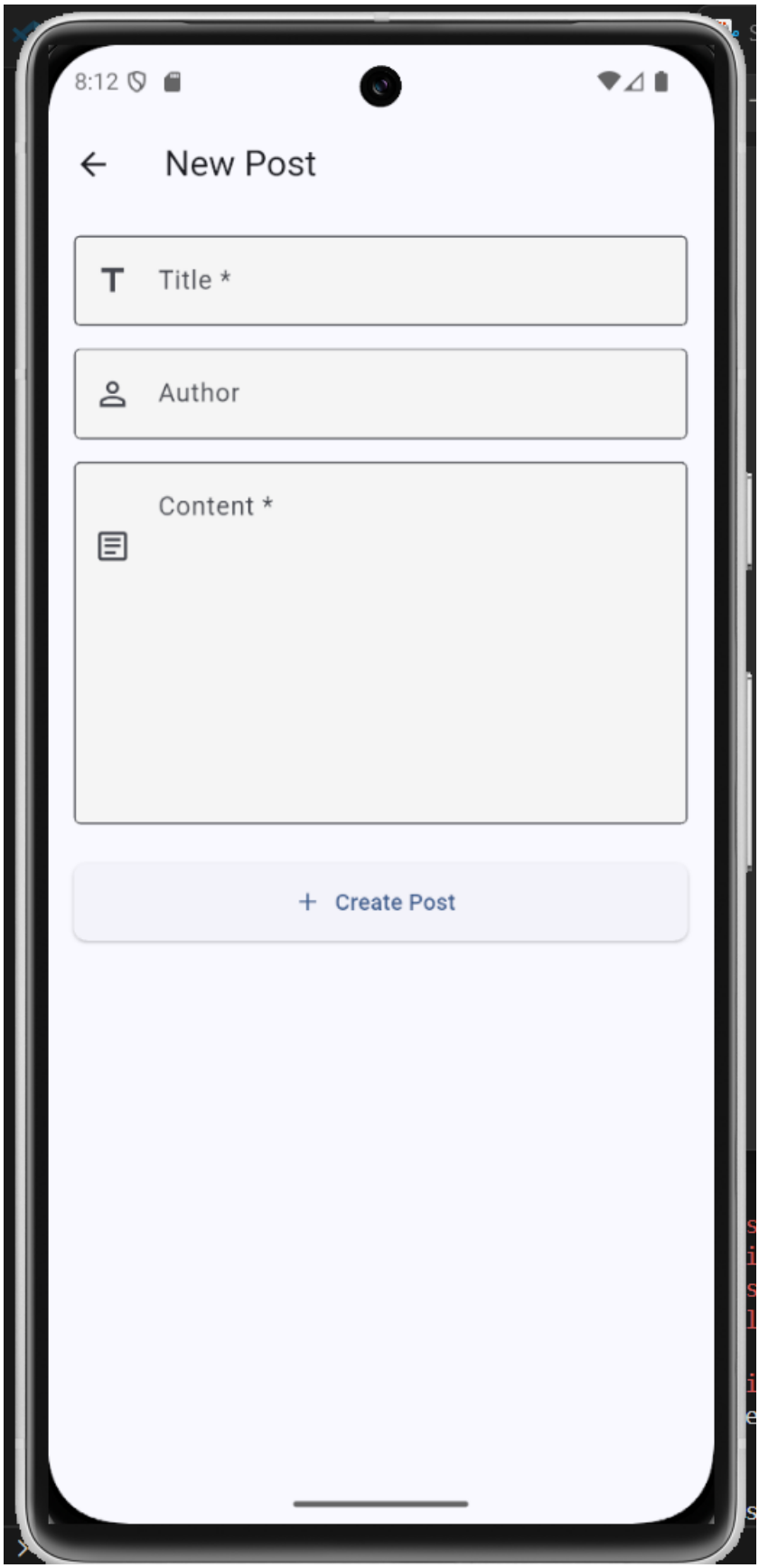
=> In DatabaseHelper, the database getter checks whether the connection is open before returning it:

```
Future<Database> get database async {
if (database != null && database!.open) return database!;
database = await _initDatabase();
return database!;
}
```

=> If _initDatabase() throws, the error is caught and re-thrown as DatabaseException with a human-readable message. The UI catches DatabaseException and shows a snackbar or error widget, preventing an unhandled crash.


8:12
Offline Posts Manager
Getting started with flutter
Flutter is google UI toolkit for builiding beautiful, natively compiled
vinner
4
HERE WE GO!
HGFSGFGDJFFD DSDFNBSFHYUEWUEW GHS DHJGSUYTEWWE BCGHDSJUYEWT JGFUYRIYHFSF...
MOISE
1
Welcome to Offline Posts Manager
This app works completely offline using SQLite. You can create, read, update and delete posts without an...
Admin
5
How SQLite Works in Flutter
SQLite stores data in a single file on the device. The sqflite package provides an async API so database c...
Dev Team B
9
New Post

CamScanner



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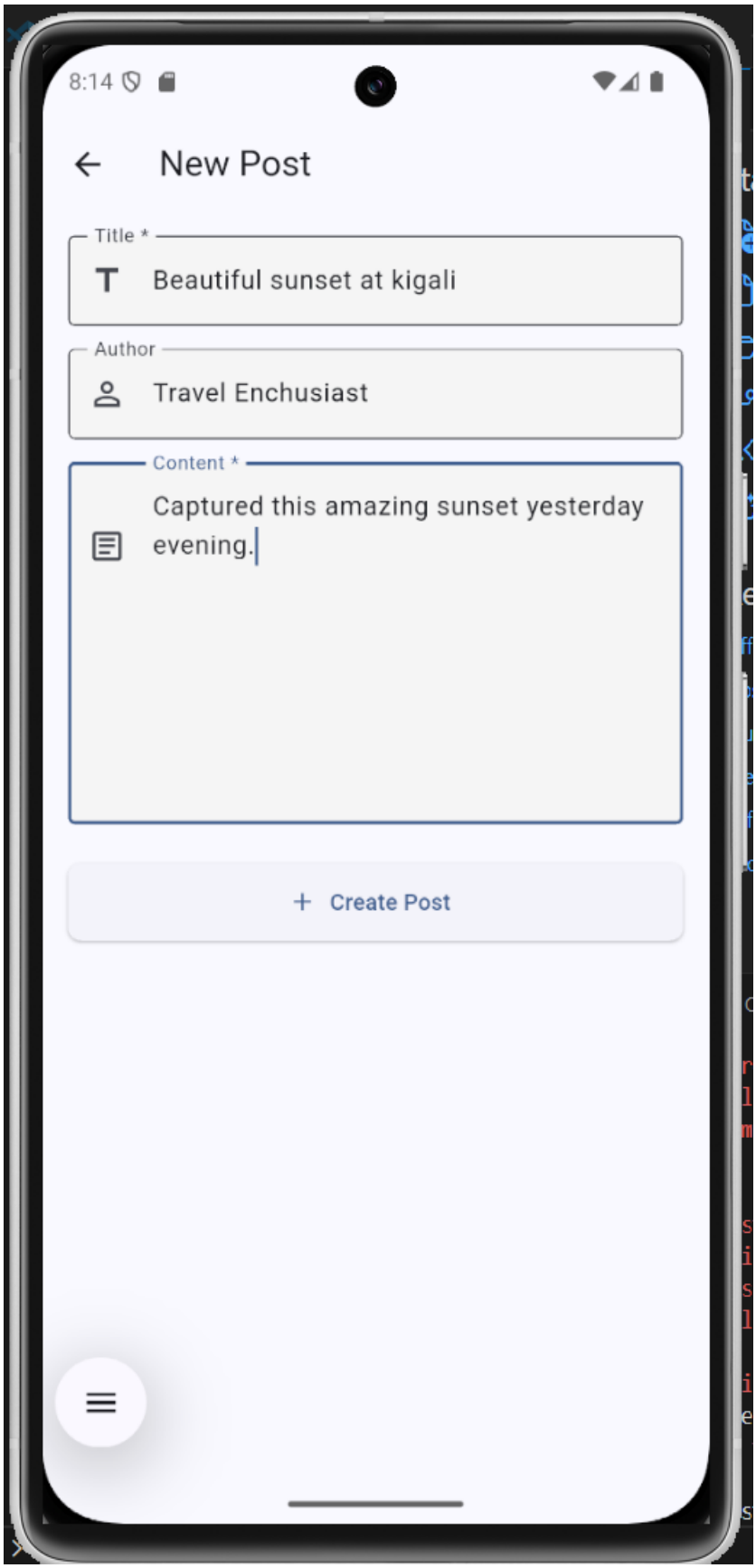
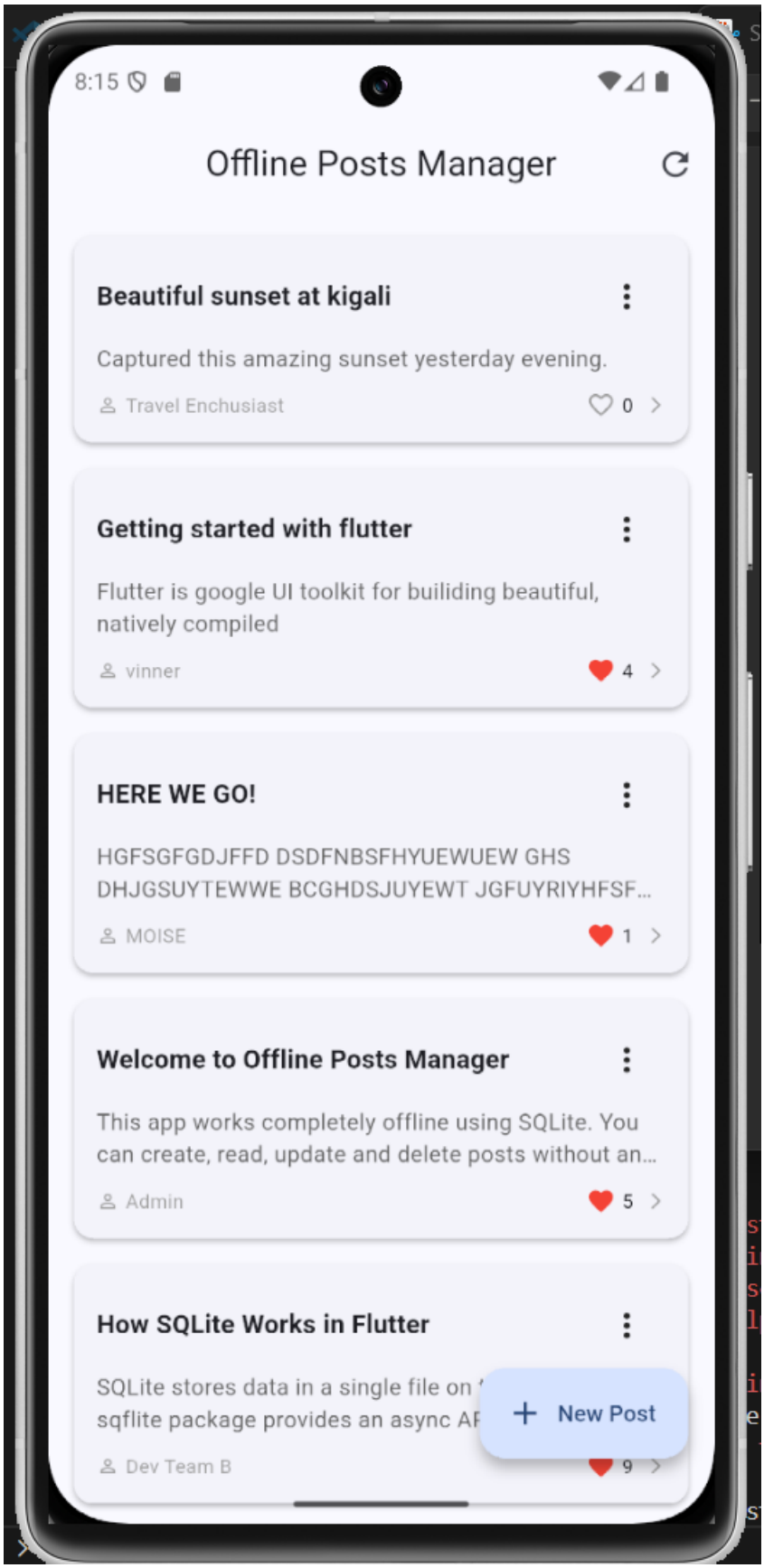

8:14
New Post
Title *
Beautiful sunset at kigali
Author
Travel Enchusiast
Content *
Captured this amazing sunset yesterday evening.
Create Post

CamScanner


8:15
Offline Posts Manager
Beautiful sunset at kigali
Captured this amazing sunset yesterday evening.
Travel Enchusiast
0
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New Post
Dev Team B
9

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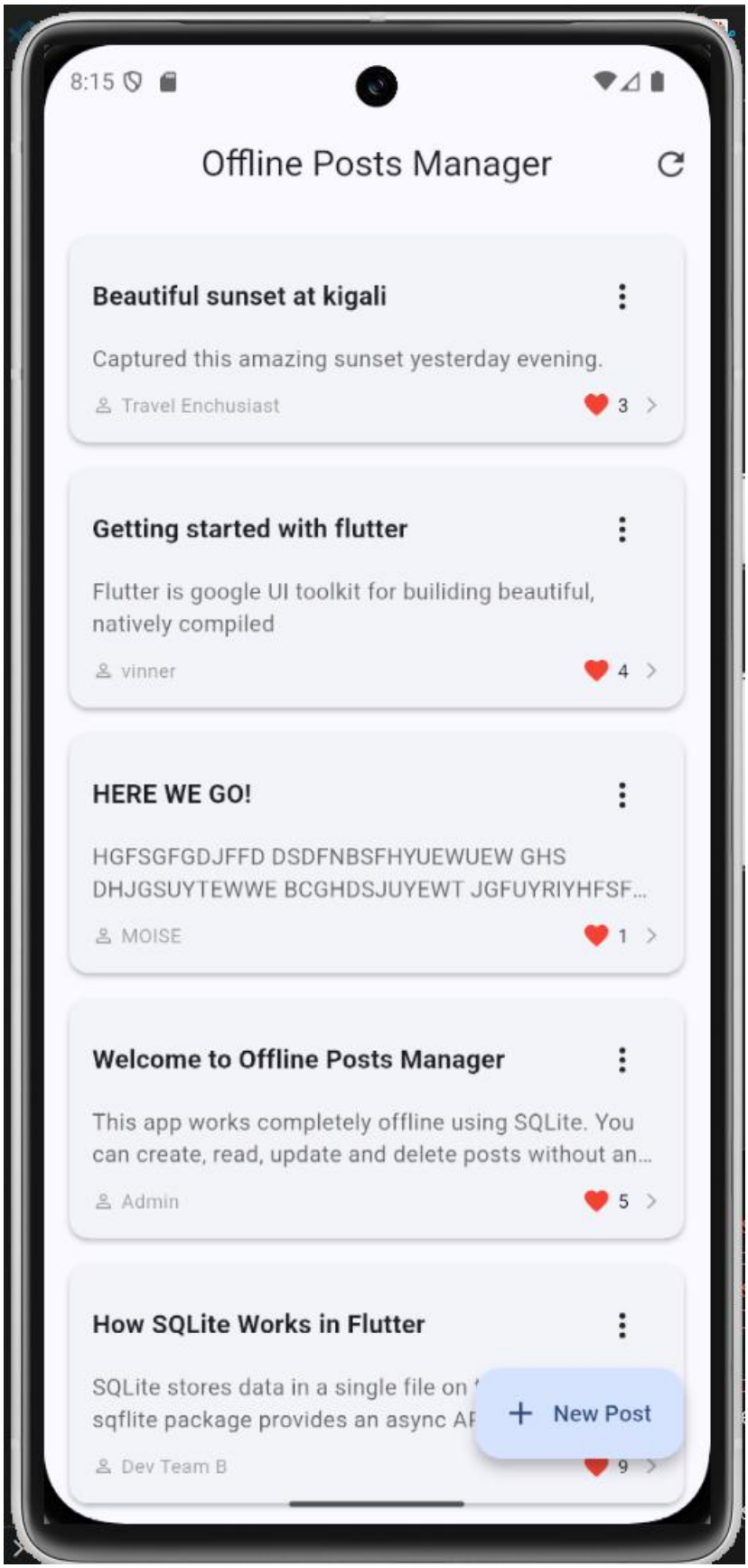
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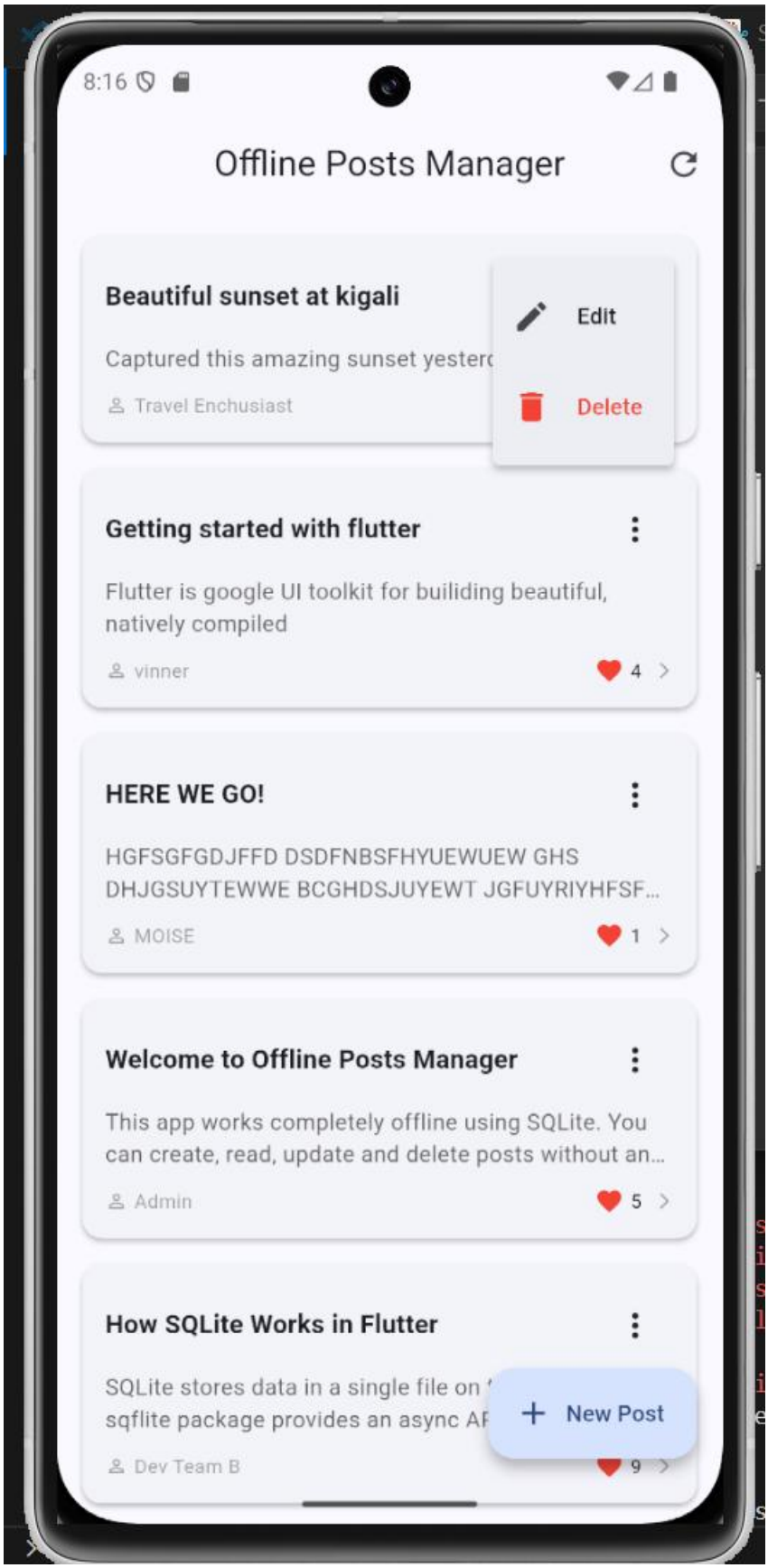
8:16
Offline Posts Manager
Beautiful sunset at kigali
Captured this amazing sunset yesterd
Travel Enchusiast
Edit
Delete
Getting started with flutter
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HERE WE GO!
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New Post

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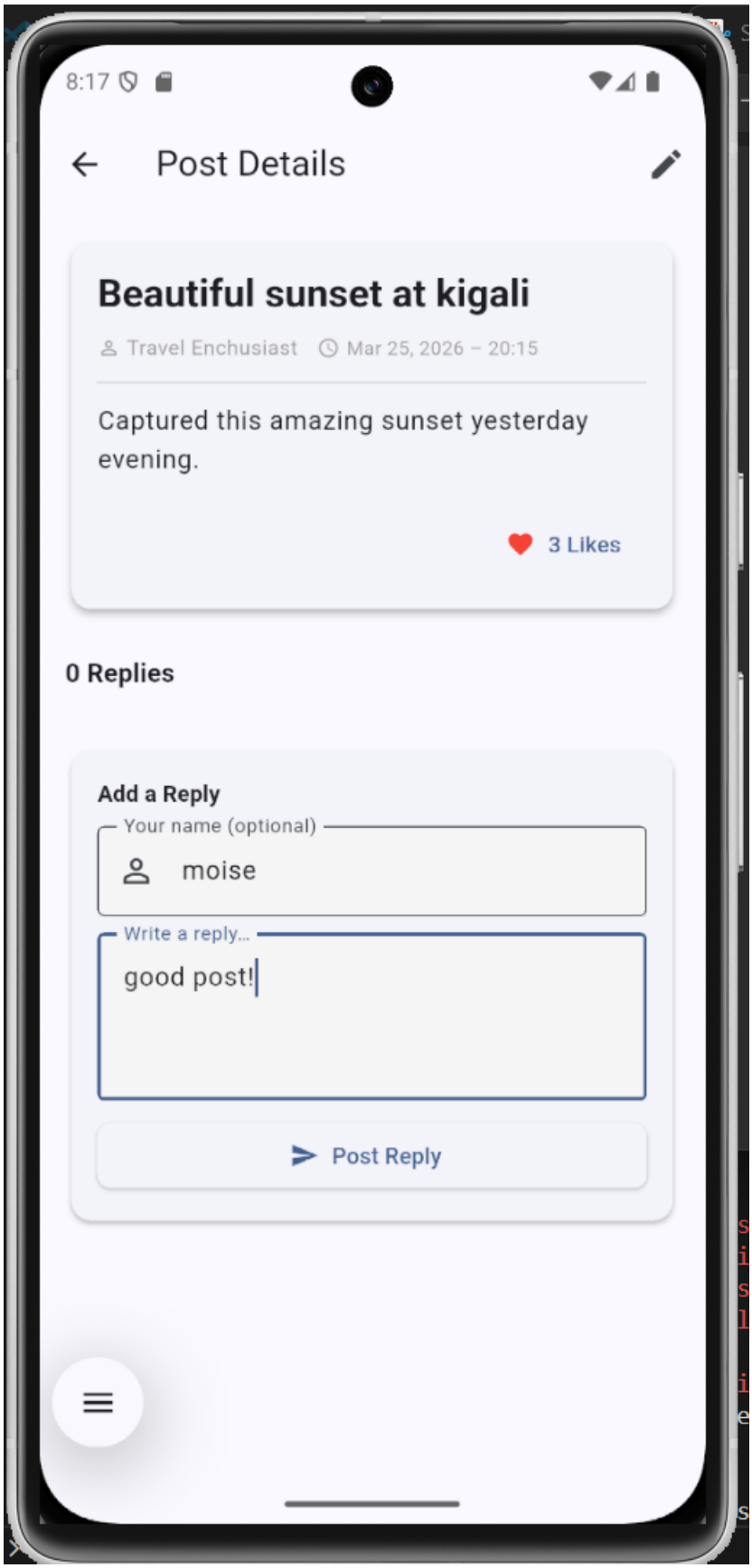
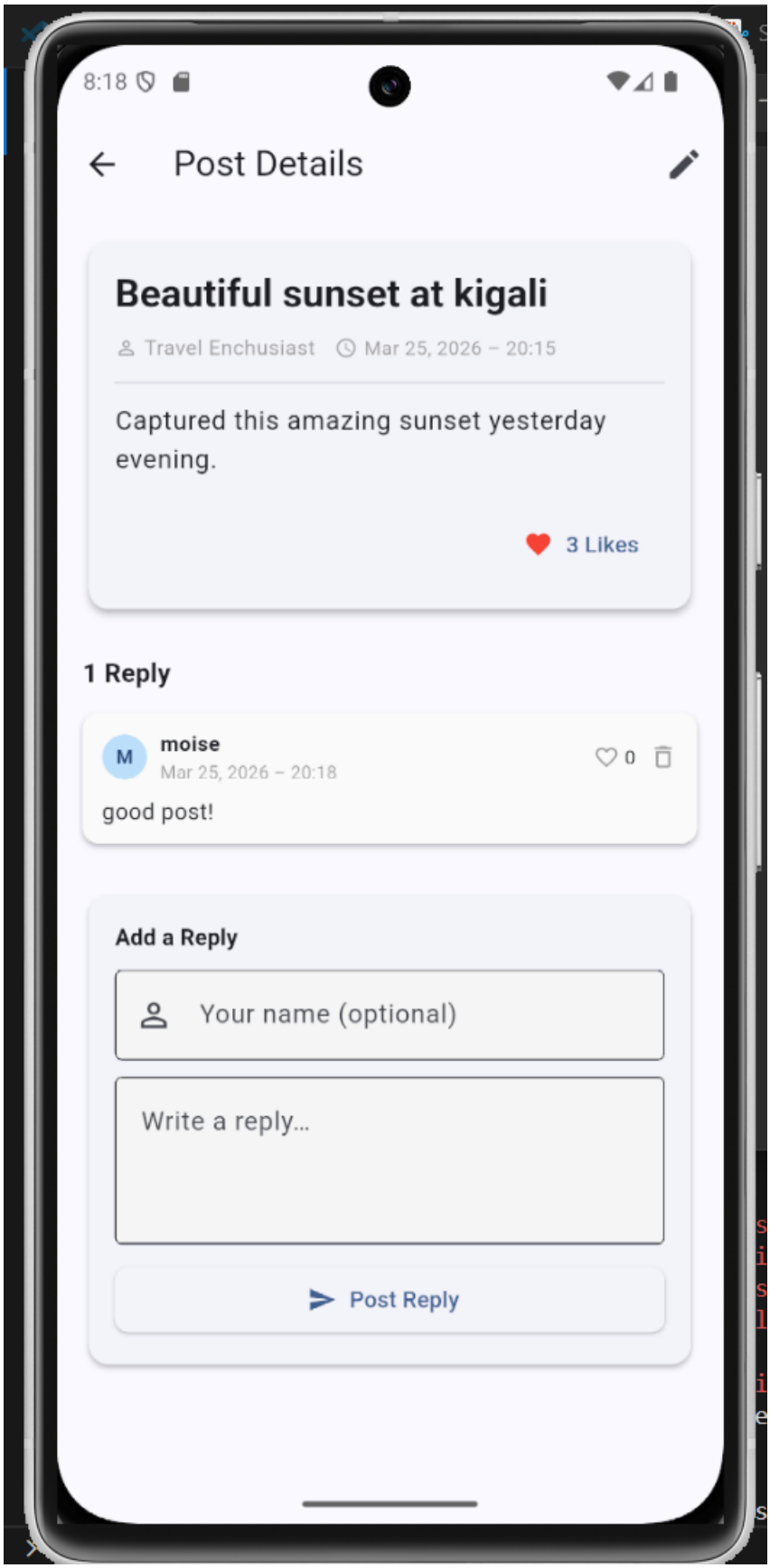
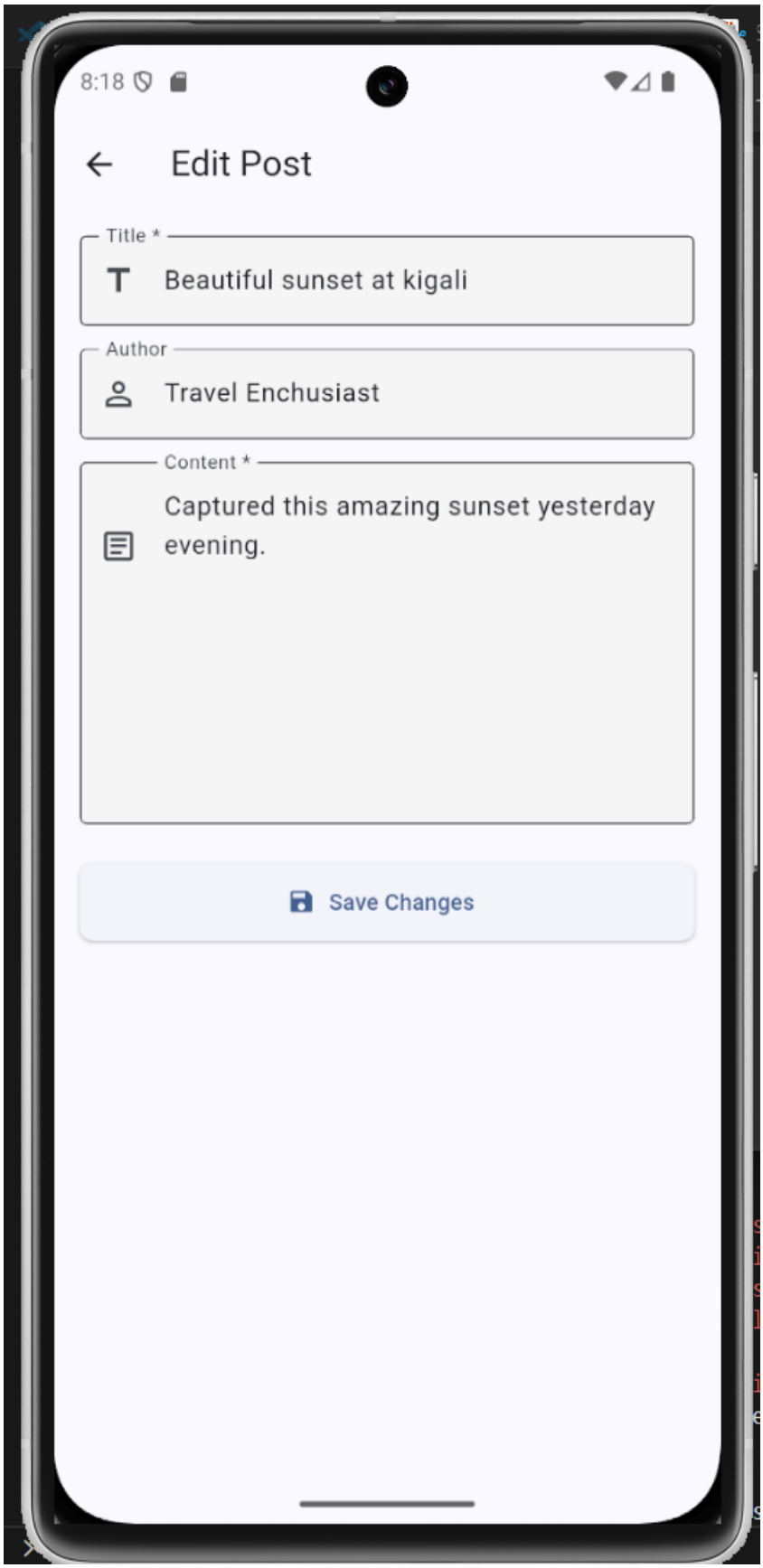

8:17
Post Details
Beautiful sunset at kigali
Travel Enchusiast
Mar 25, 2026 – 20:15
Captured this amazing sunset yesterday evening.
3 Likes
0 Replies
Add a Reply
Your name (optional)
moise
Write a reply...
good post!
Post Reply

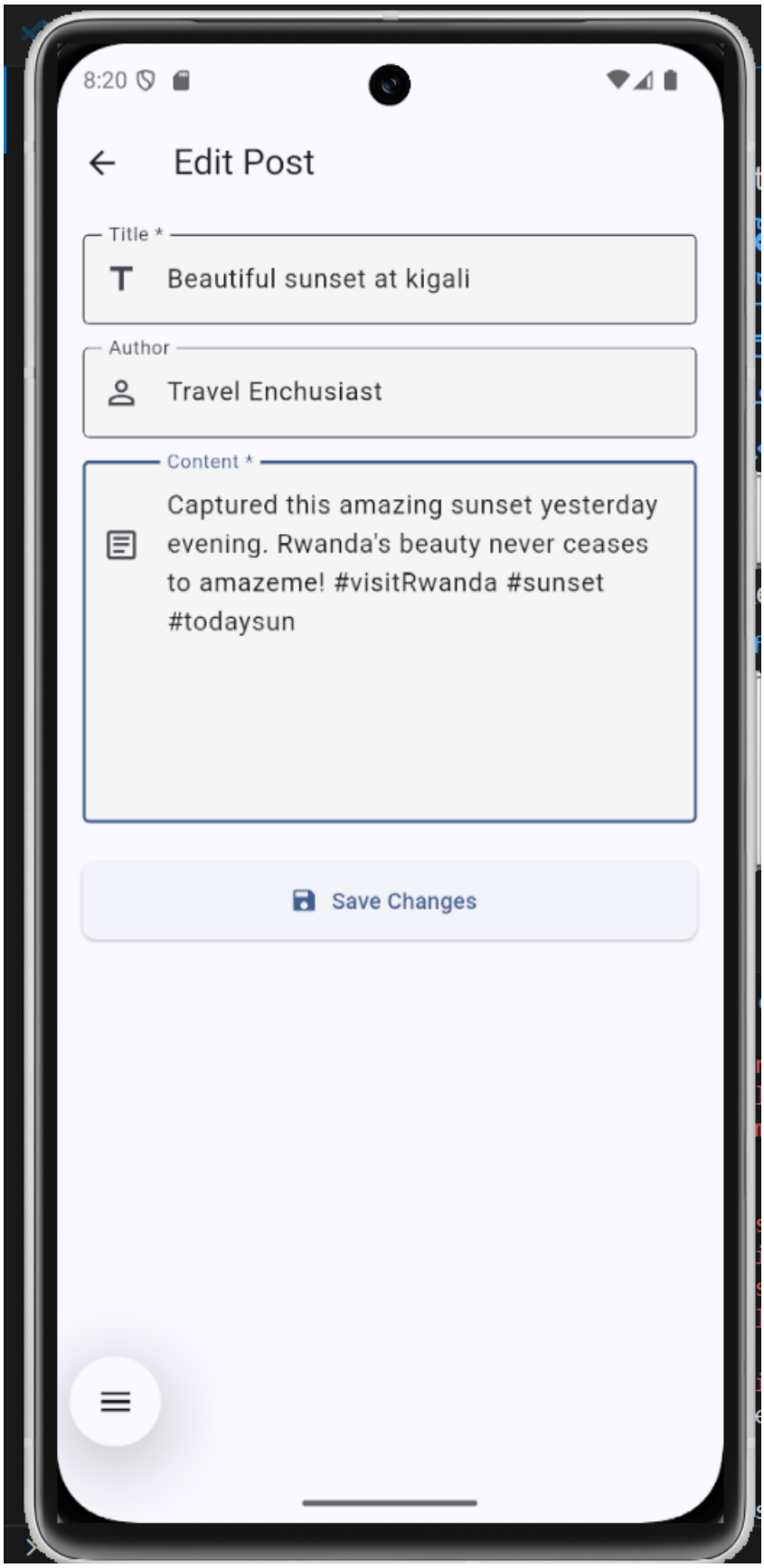
CamScanner


8:18
Post Details
Beautiful sunset at kigali
Travel Enchusiast
Mar 25, 2026 – 20:15
Captured this amazing sunset yesterday evening.
3 Likes
1 Reply
M
moise
Mar 25, 2026 – 20:18
0
good post!
Add a Reply
Your name (optional)
Write a reply...
Post Reply

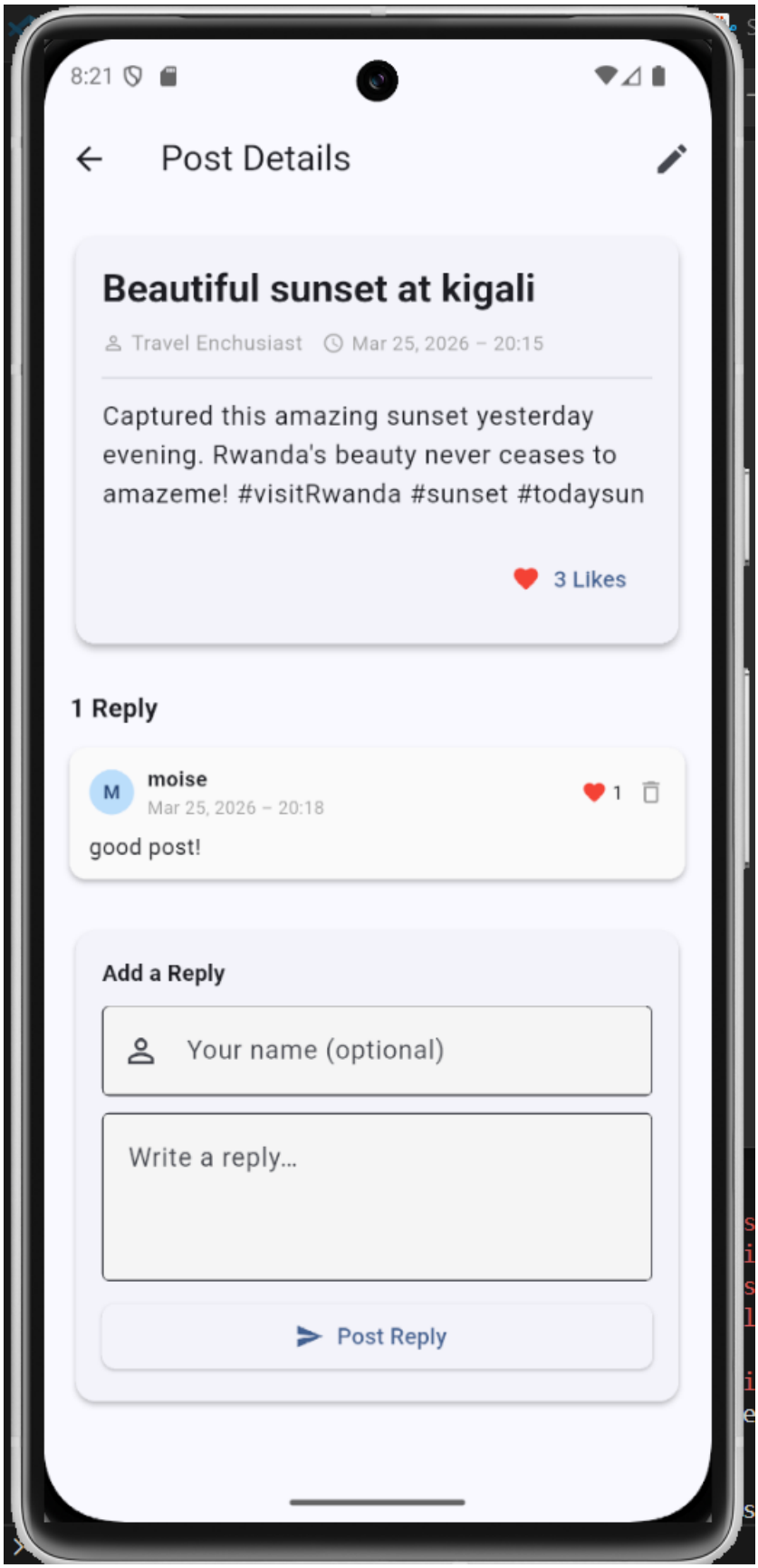
CamScanner


8:18
Edit Post
Title *
Beautiful sunset at kigali
Author
Travel Enchusiast
Content *
Captured this amazing sunset yesterday evening.
Save Changes

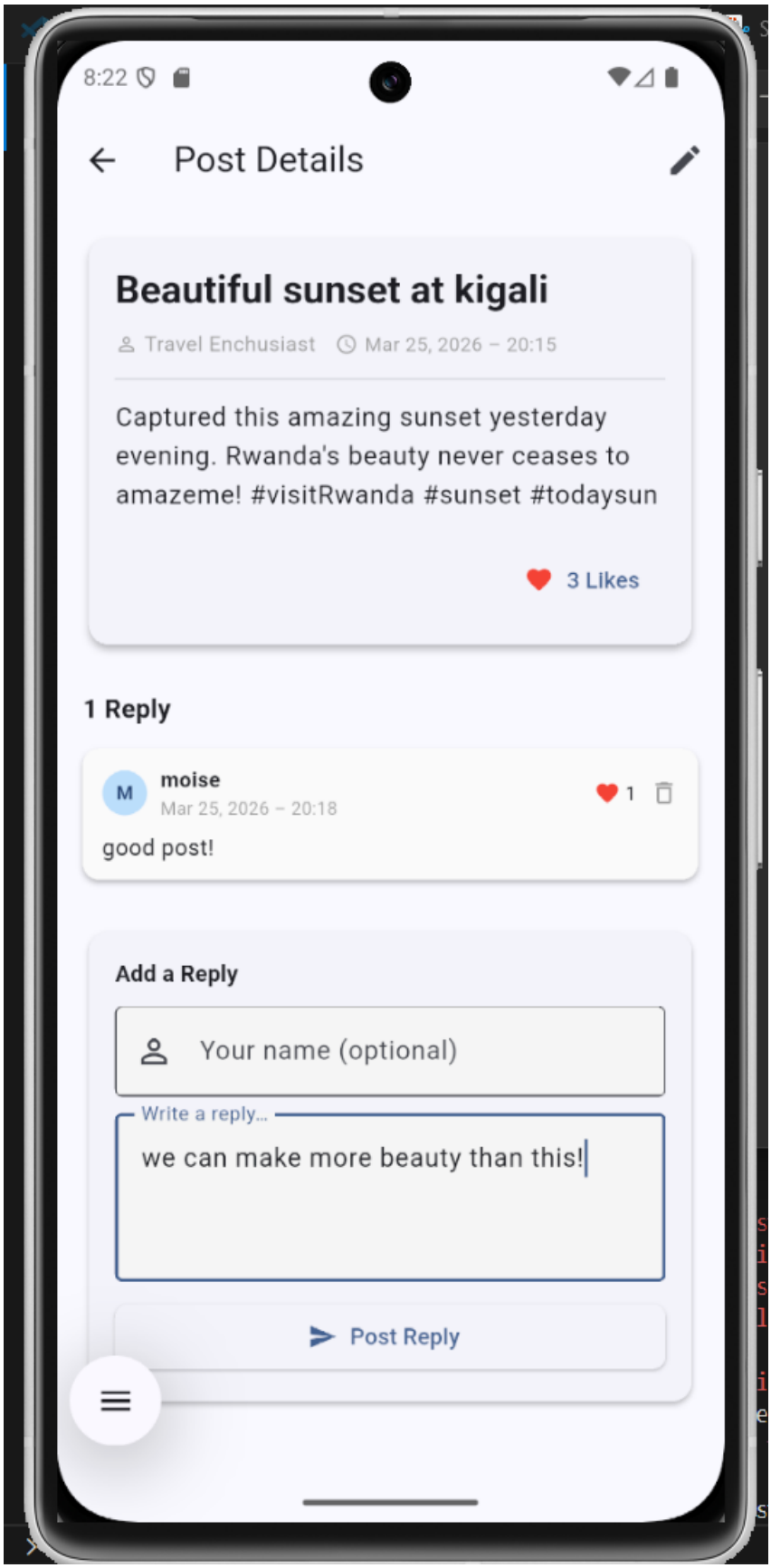
CamScanner


8:20
Edit Post
Title *
Beautiful sunset at kigali
Author
Travel Enchusiast
Content *
Captured this amazing sunset yesterday evening. Rwanda's beauty never ceases to amazeme! #visitRwanda #sunset #todaysun
Save Changes

CamScanner


8:21
Post Details
Beautiful sunset at kigali
Travel Enchusiast
Mar 25, 2026 – 20:15
Captured this amazing sunset yesterday evening. Rwanda's beauty never ceases to amazeme! #visitRwanda #sunset #todaysun
3 Likes
1 Reply
M
moise
Mar 25, 2026 – 20:18
1
good post!
Add a Reply
Your name (optional)
Write a reply...
Post Reply

CamScanner


8:22
Post Details
Beautiful sunset at kigali
Travel Enchusiast
Mar 25, 2026 – 20:15
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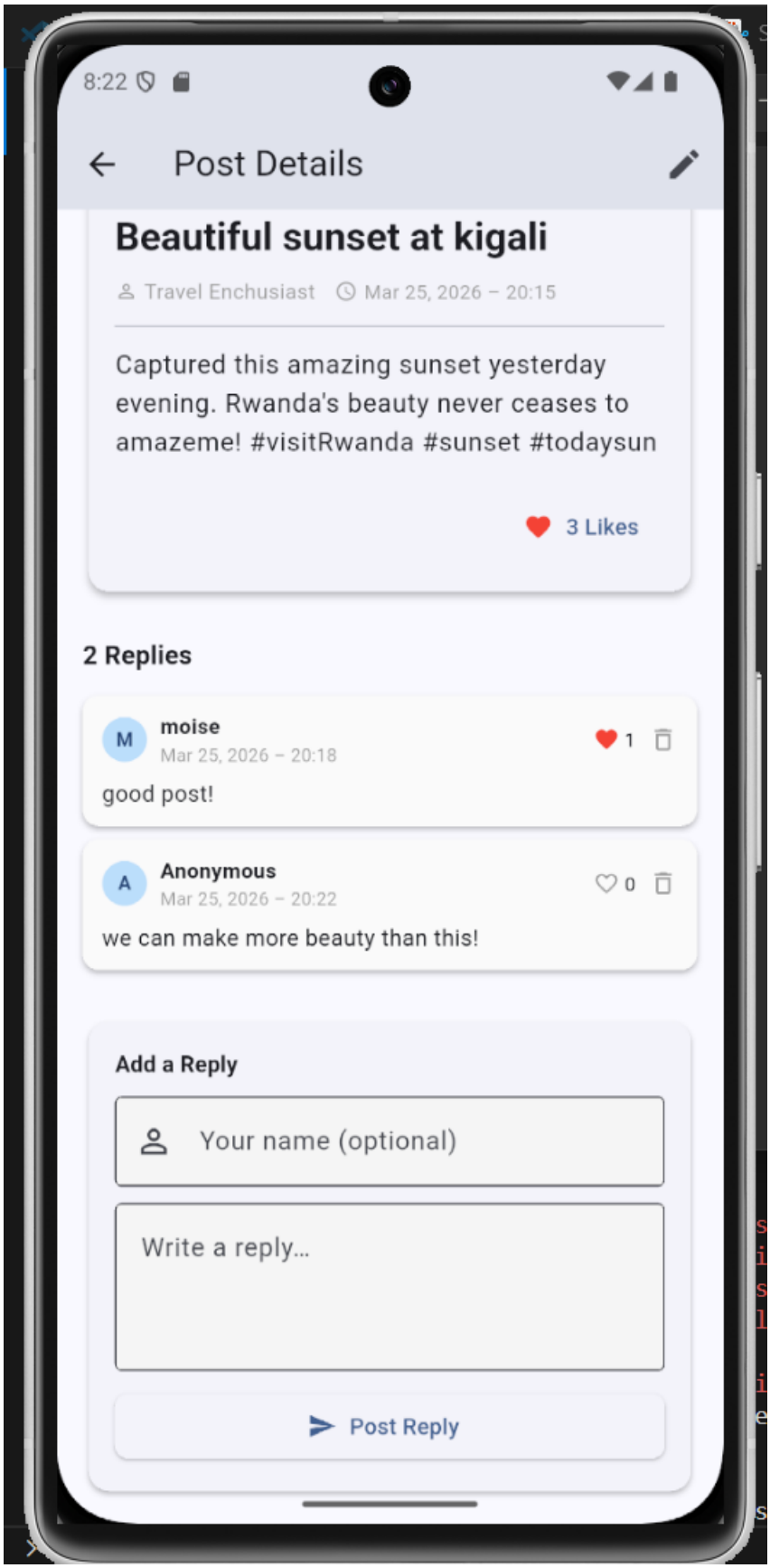
3 Likes
1 Reply
M
moise
Mar 25, 2026 – 20:18
1
good post!
Add a Reply
Your name (optional)
Write a reply...
we can make more beauty than this!
Post Reply

CamScanner


8:22
Post Details
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Travel Enchusiast
Mar 25, 2026 – 20:15
Captured this amazing sunset yesterday evening. Rwanda's beauty never ceases to amazeme! #visitRwanda #sunset #todaysun
3 Likes
2 Replies
M
moise
Mar 25, 2026 – 20:18
1
good post!
A
Anonymous
Mar 25, 2026 – 20:22
0
we can make more beauty than this!
Add a Reply
Your name (optional)
Write a reply...
Post Reply

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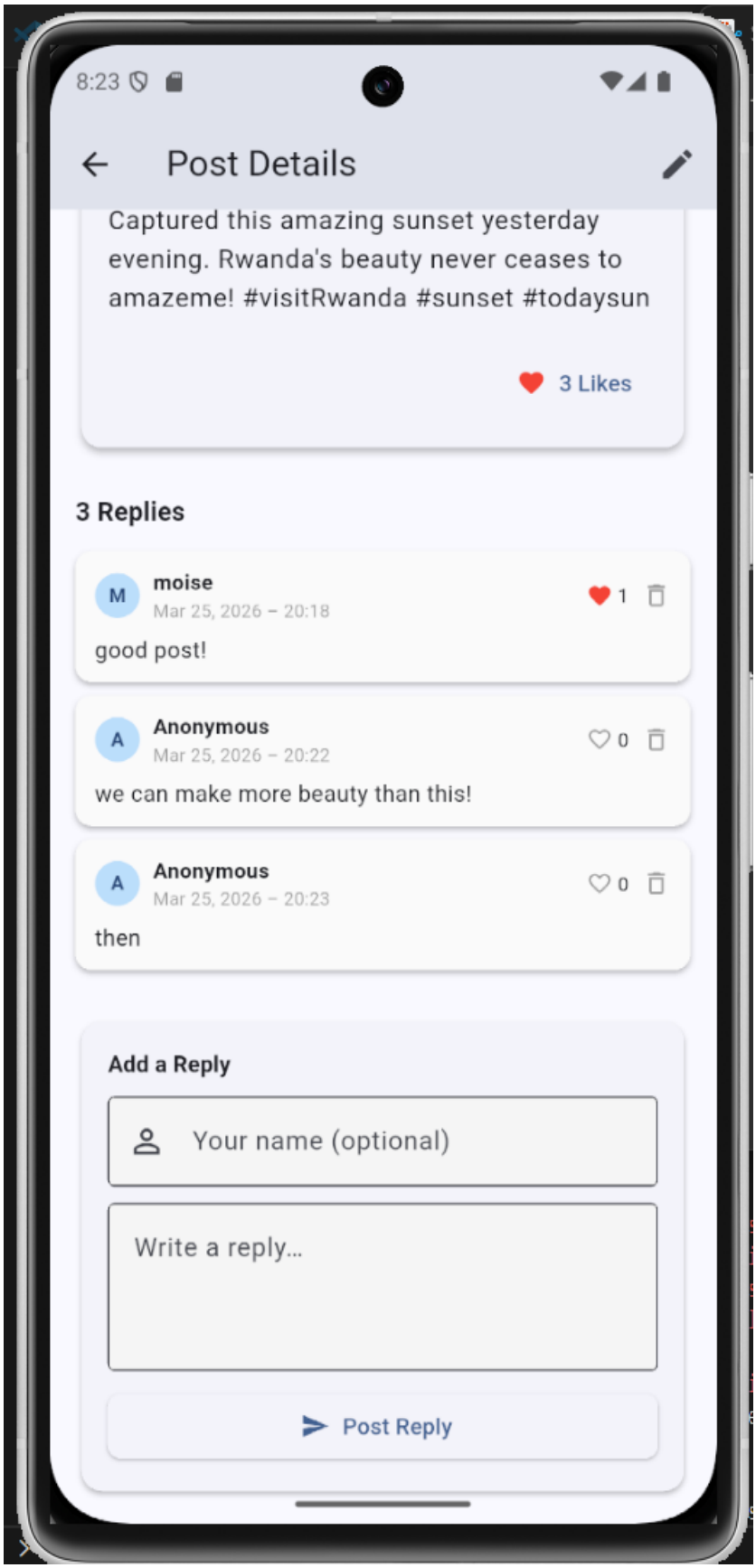
8:23
Post Details
Captured this amazing sunset yesterday evening. Rwanda's beauty never ceases to amazeme! #visitRwanda #sunset #todaysun
3 Likes
3 Replies
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Mar 25, 2026 – 20:18
1
good post!
A
Anonymous
Mar 25, 2026 – 20:22
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we can make more beauty than this!
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Anonymous
Mar 25, 2026 – 20:23
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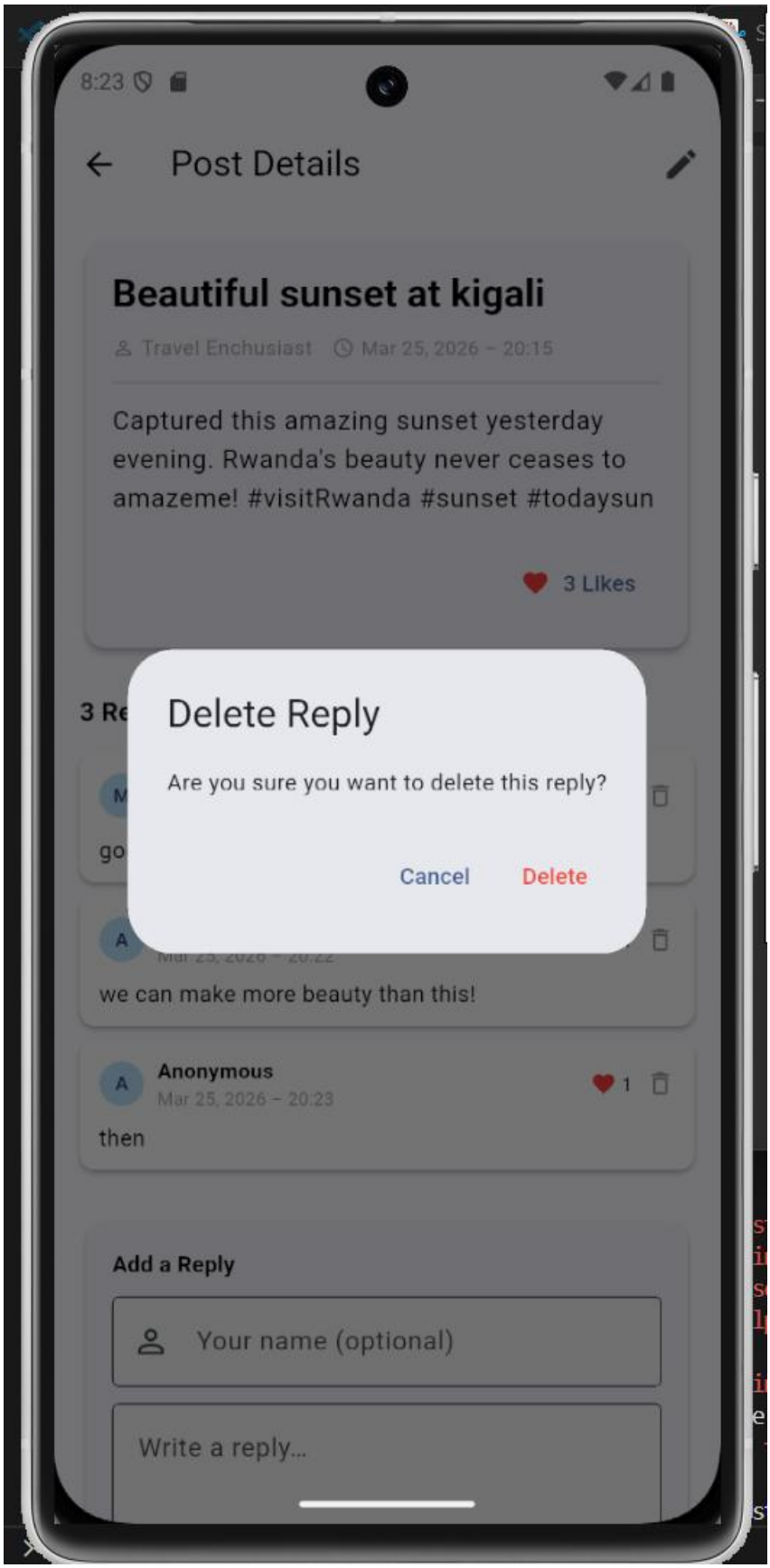
then
Add a Reply
Your name (optional)
Write a reply...
Post Reply

CamScanner


8:23
Post Details
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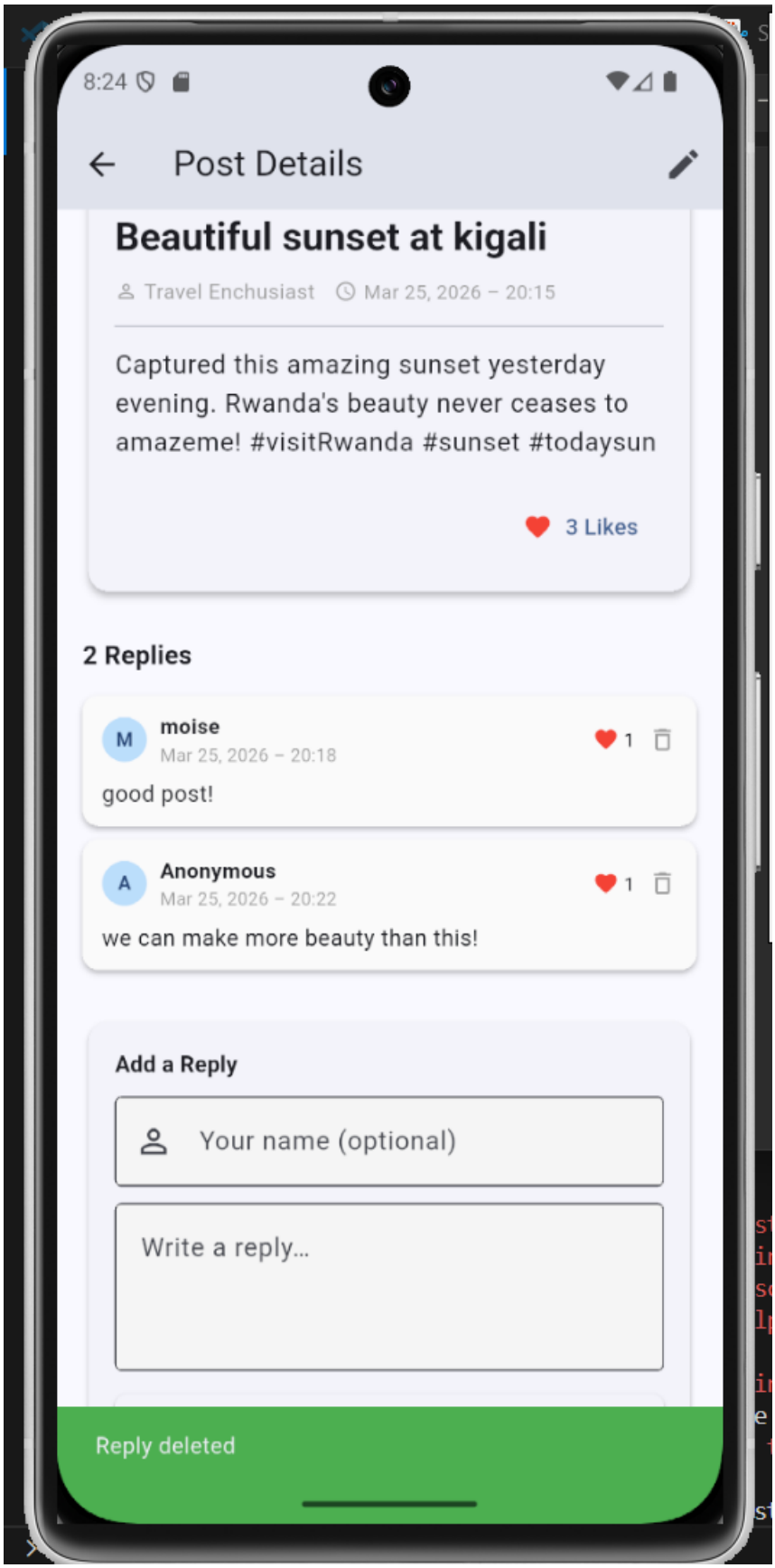
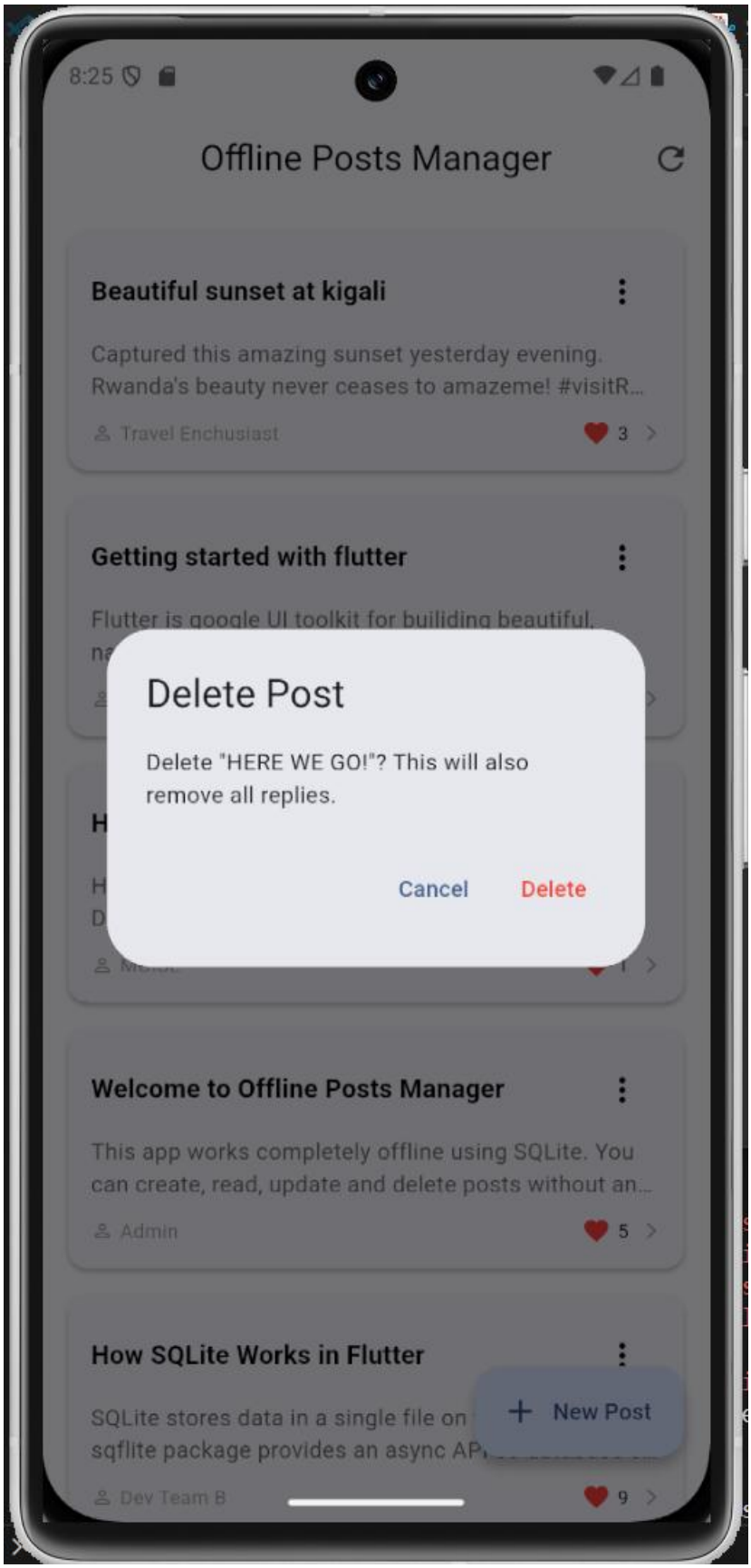
Travel Enchusiast
Mar 25, 2026 – 20:15
Captured this amazing sunset yesterday evening. Rwanda's beauty never ceases to amazeme! #visitRwanda #sunset #todaysun
3 Likes
Delete Reply
Are you sure you want to delete this reply?
Cancel
Delete
we can make more beauty than this!
Anonymous
Mar 25, 2026 – 20:23
1
then
Add a Reply
Your name (optional)
Write a reply...

CamScanner


8:24
Post Details
Beautiful sunset at kigali
Travel Enchusiast
Mar 25, 2026 – 20:15
Captured this amazing sunset yesterday evening. Rwanda's beauty never ceases to amazeme! #visitRwanda #sunset #todaysun
3 Likes
2 Replies
M
moise
Mar 25, 2026 – 20:18
1
good post!
A
Anonymous
Mar 25, 2026 – 20:22
1
we can make more beauty than this!
Add a Reply
Your name (optional)
Write a reply...
Reply deleted

CamScanner


8:25
Offline Posts Manager
Beautiful sunset at kigali
Captured this amazing sunset yesterday evening. Rwanda's beauty never ceases to amazeme! #visitR...
Travel Enchusiast
3
Getting started with flutter
Flutter is google UI toolkit for builiding beautiful,
Delete Post
Delete "HERE WE GO!"? This will also remove all replies.
Cancel
Delete
Welcome to Offline Posts Manager
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Admin
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How SQLite Works in Flutter
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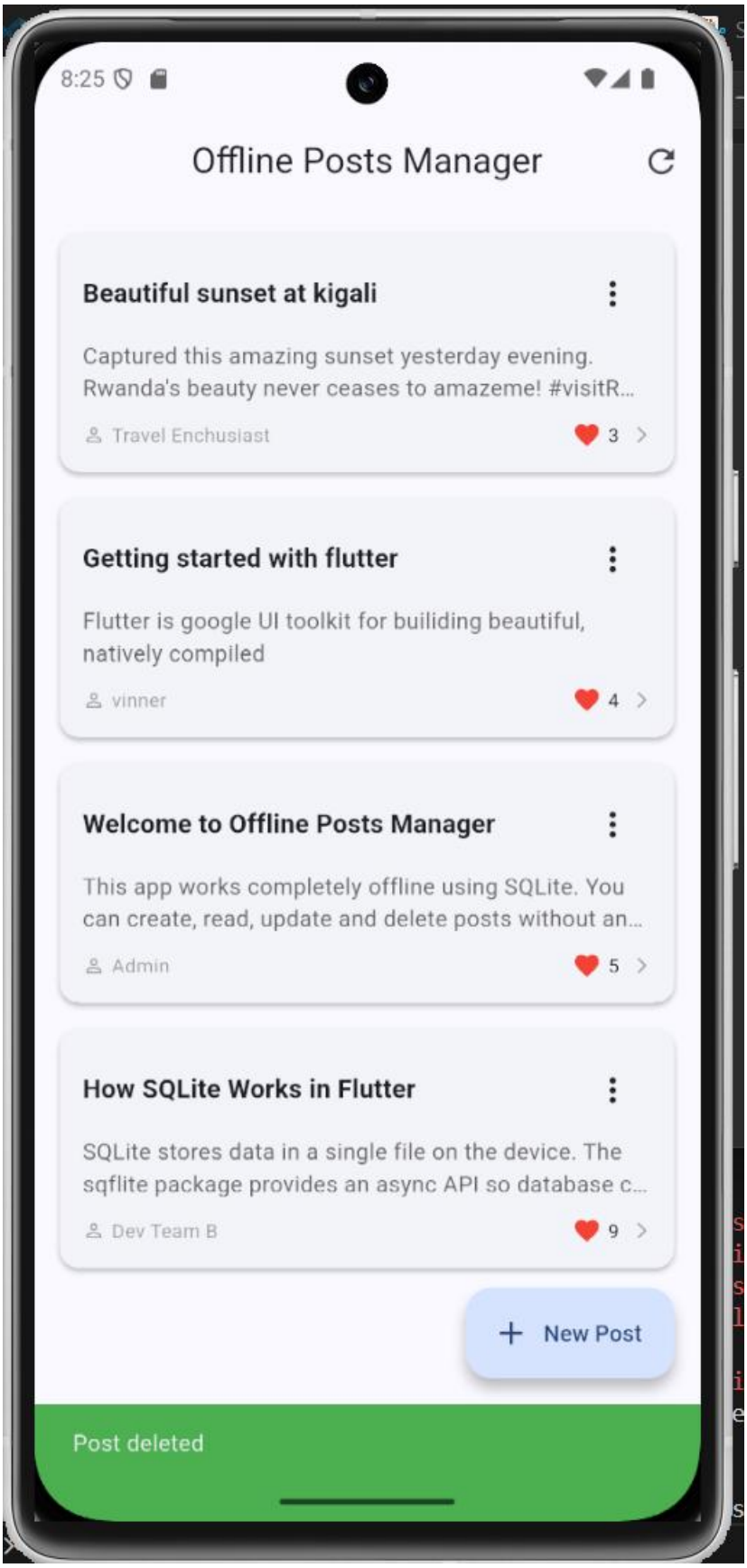
New Post
Dev Team B
9

CamScanner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Dev Team B
9
New Post
Post deleted

CamScanner

Q.2.2. Handle Insert / update / Delete Errors.

- Every CRUD method is wrapped in a try / catch block.
- On failure the raw sqflite error is wrapped in custom DatabaseException so the UI layer only needs to catch one type.
- Conflict Algorithm replace is used on inserts to gracefully handle duplicate-id conflicts.
- The updatePost method checks that post.id != null before attempting an update preventing a silent no-op.

Q 2.3. Handle invalid or corrupted Data.

- post.fromMap and reply.fromMap use null-coalescing(??) on every field, falling back to safe defaults instead of throwing a type-cast error.
- DataTime.parse is called inside a factory constructor. If the stored string is malformed DataTime.now() is used as a fallback.
- validation helpers (_validatePost, _validateReply) check for empty string before any DB call, surfacing the error to the user immediately rather than writing invalid data.

Q3. Explain sqlite in flutter.

3.1. Database vs Table.

=> Database is the container - in Flutter / sqflite. it is a single .db file stored on the device. It can hold many tables.

=> A Table is a two-dimensional structure of rows and columns inside the Database, similar to a spreadsheet. In this app there Two tables:

- posts : stores title, body, author, likes, created_at, updated_at
- replies: stores postid (foriegn key), body, likes, created_at.

Q3.2. CRUD operations.

Operation	SQFlite Method	Description.
Create	db.insert()	inserts a new Map <String, dynamic> row and returns the new auto-increment id
Read	db.query()	Returns a list <Map> of rows; supports where, orderBy, limit.
Update	db.update()	modifies rows matching a where clause; returns number of affected rows.
Delete	db.delete()	Removes rows matching a where clause; returns number of deleted rows

Q3.3. How Flutter interacts with the Database Asynchronously.

=> SQlite file I/O can be slow (milliseconds) which would freeze a 60fps UI if run on the main Thread. Sqflite solves this by running all database calls on background isolate and returning Dart Futures:

- Every DB method is declared async and awaited by the caller
- The UI never block - while data loads, a Circular Progress Indicator is shown
- setstate() is called only after the future completes